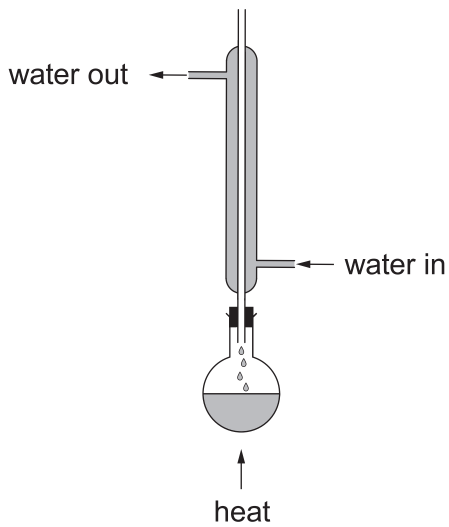
Examiner only

9. *(a)* Alcohols react with carboxylic acids to form esters. To prepare a pure sample of an ester a condenser can be used in the first stage.



(i) Name the method being used with the condenser positioned in this way. Explain why it is necessary to use the condenser. [2]

....

....

....

(ii) Name the method used to separate the product from the liquid mixture. State which property of an ester allows this method of separation to be used. [2]

Method ....

Property ....

(iii) Name the catalyst most commonly used in the preparation of an ester. [1]

....

(iv) Write the equation for the reaction between methanoic acid and butan-2-ol. Clearly show the structure of the ester formed. [2]

Examiner only

(b) A compound is known to be one of butan-2-ol, $CH_3CH_2CH(OH)CH_3$, 2-methylpropanoic acid, $CH_3CH(CH_3)COOH$, and 3-hydroxybutanoic acid, $CH_3CH(OH)CH_2COOH$.

Choose **two** chemical tests that will enable you to determine which one it is.

Complete the table below.

Give the reagent(s) for each test and the observation expected for a positive result.

For each test put a **tick (✓)** in the box to show the compound that gives a positive result and a **cross (×)** for those that do not. [5]

Reagent(s)	Observation expected for positive result	butan-2-ol	2-methylpropanoic acid	3-hydroxybutanoic acid

12